

RS485 TO ETH (B) Connect Alibaba Cloud And EMQX

From Waveshare Wiki

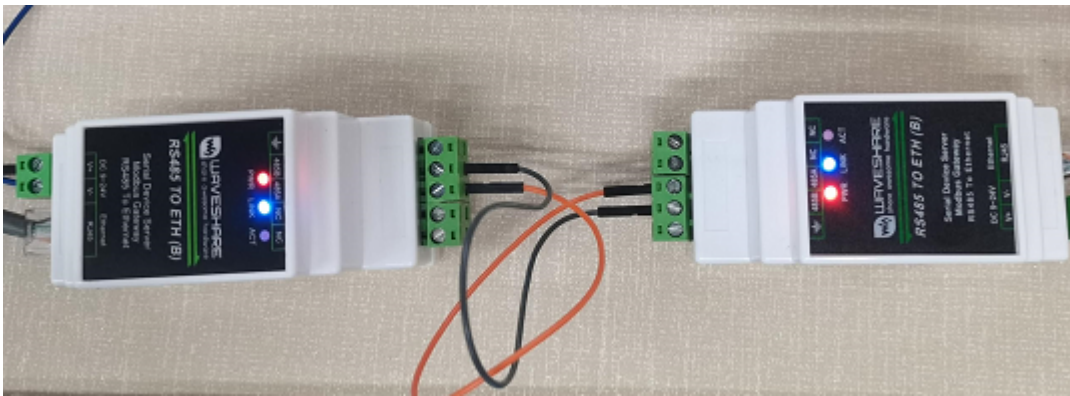
Jump to: navigation, search

Hardware Connection

Preparation

- PC x 1
- RS485 TO ETH(B) x 2 (<https://www.waveshare.com/rs485-to-eth-b.htm>)

How To Connect



(/wiki/File:RS485_TO_ETH_(B)_Connect01.png)

Software Preparation

- Alibaba Cloud (<https://account.alibabacloud.com/login/login.htm?spm=5176.12901015-2.0.0.3dd3525cwUtaAb>)
- EMQX Cloud (https://www.emqx.com/en/cloud?utm_source=baidu&utm_medium=cp&utm_campaign=search-cloud-brand)
- VirCOM (<https://files.waveshare.com/upload/0/0a/VirCom.rar>)

Start Communication

- Open Vircom->Device Management-> Double-click the corresponding device -> Firmware Config-> MQTT Config.

Webpage directly download mode

Webpage directly in local PC:

E:\FAQ-QUECTEL\485 TO ETH\RS485 TO ETH B\202241MQTT

Special configs:

Clear all

ZLMB config **MQTT config** JSON config Reg packet

Code file download mode

Select code file:

C:\firmware.bin

Download through the network

Device IP address or domain: 192.168.10.61

Download port (Don't modify): 1092

Download through serial port

Serial port: COM1

Baudrate: 115200

Device modual/type: 2003

Flash size: 256 KB

DevID: 285FCAD56BAD Bind ID

Please close the opened webpage of the modual in the browser, before start download.

Download

(/wiki/File:RS485_TO_ETH_(B)_Connect02.png)

- Alibaba cloud platform -> Device -> Device Information.

← 7600 Online

Products 7600 View

ProductKey a1mQpKOF9hp Copy

DeviceSecret ***** View

Device Information Topic List TSL Data Device Shadow Manage Files Device Log Online Debug Groups

Device Information

Product Name	7600	ProductKey	a1mQpKOF9hp Copy
Node Type	Devices		
Alias	Edit		
Created At	Dec 12, 2020		
Current Status	Online		
MQTT Connection Parameters	Here		

More Device Information

SDK Language -

MQTT Connection Parameters

clientid	a1mQpKOF9hp00[securemode=2,signmethod=hmacsha256,timestamp=1648886336647]
username	7600a1mQpKOF9hp
passwd	033620b697091ef8b89c809db851f10e27a5f915cb42fd966e
mqttHostUrl	a1mQpKOF9hp.mqtt.cn-shanghai.aliyuncs.com
port	1883

Copy Close

(/wiki/File:RS485_TO_ETH_(B)_Connect03.png)

- Configure the MQTT parameters of the Alibaba Cloud IoT platform to one of the RS485 TO ETH (B).

MQTT connect settings

MQTT server IP:	47.103.184.100
MQTT server port:	1883
User name:	7600&a1mQpKOF9hp
Key:	*****
MQTT ID:	7600 securemode=3,signmethod=hmacs
Subscription topic:	/a1mQpKOF9hp/7600/user/7600
Publish topic:	/a1mQpKOF9hp/7600/user/7600

Advanced Save Delete Cancel

(/wiki/File:RS485_TO_ETH_(B)_Connect04.png)

- Download MQTT parameters to RS485 TO ETH(B).

☒ Webpage directly download mode

Webpage directly in local PC:

E:\FAQ-QUECTEL485 TO ETH\RS485 TO ETH B\202241MQTT

Special configs:

☐ Code file download mode

Select code file:

C:\firmware.bin

☒ Download through the network

Device IP address or domain: 192.168.10.61

Download port (Don't modify): 1092

☐ Download through serial port

Serial port: COM1

Baudrate: 115200

Device modual/type: 2003

Flash size: 256 KB

DevID: 285FCAD56BAD

Please close the opened webpage of the modual in the browser, before start download.



(/wiki/File:RS485_TO_ETH_(B)_Connect05.png)

- Check the parameter of MQTT text in the corresponding file.

:) > FAQ-QUECTEL > RS485 TO ETH B > MQTT

```

mqtt.txt - Notepad
File Edit Format View Help
[MQTT]
VER=4
CLIENT_ID=a1mQpKOF9hp.7600|securemode=2,signmethod=hmacsha256,timestamp=1648880965464|
USERNAME=7600&a1mQpKOF9hp
PASSWD=1d33af6d3d5968b61732f0d1d8bbcd3148d2e83f75ca2ec6c44d093fc6ccd328
KEEPALIVE=60
CLEAN_SESSION=1
WILL_FLAG=0
WILL_TOPIC=
WILL_MESSAGE=
WILL_RETAIN=0
WILL_QOS=0
SUBSCRIBE_TOPIC=/a1mQpKOF9hp/7600/user/7600
SUBSCRIBE_QOS=1
PUBLISH_TOPIC=/a1mQpKOF9hp/7600/user/7600
PUBLISH_QOS=1
PUBLISH_RETAIN=0

```

(/wiki/File:RS485_TO_ETH_(B)_Connect06.png)

- Config MQTT parameter -> The Link light of RS485 TO ETH changes from yellow to blue -> successfully connected to the MQTT server

Device Settings

Device Info	Network	Advanced Settings
Virtual Serial: Not Use	IP Mode: Static	DNS Server IP: 202 . 96 . 134 . 33
Dev Type:	IP Address: 192 . 168 . 10 . 61	Dest. Mode: Dynamic
Dev Name: WSDEV0001	Port: 0	Transfer Protocol: None
Dev ID: 2B5FCAD56BAD [-]	Work Mode: TCP Client	Keep Alive Time: 60 (s)
Firmware Ver: V1.452	Net Mask: 255 . 255 . 252 . 0	Reconnect Time: 12 (s)
	Gateway: 192 . 168 . 11 . 1	Http Port: 80
	Dest. IP/Domain: 47.103.184.100 Local IP	UDP Group IP: 230 . 90 . 76 . 1
	Dest. Port: 1883	☐ Register Pkt: ☐ ASCII
		☐ Restart for no data every 300 Sec.
		☐ Enable send parameter every 5 Min.
		More Advanced Settings...
		Framing Rule
		Max Frame Length: 1300 (Byte)
		Max Interval(Smaller will better): 3 (Ms)

Function of the device

- ☐ Web Download
- ☒ DNS System
- ☒ REAL_COM Protocol
- ☒ Modbus TCP To RTU
- ☒ Serial Commnad
- ☒ DHCP Support
- ☐ Storage Extend
- ☒ Multi-TCP Connection

Serial

Baud Rate: 115200

Data Bits: 8

Parity: None

Stop Bits: 1

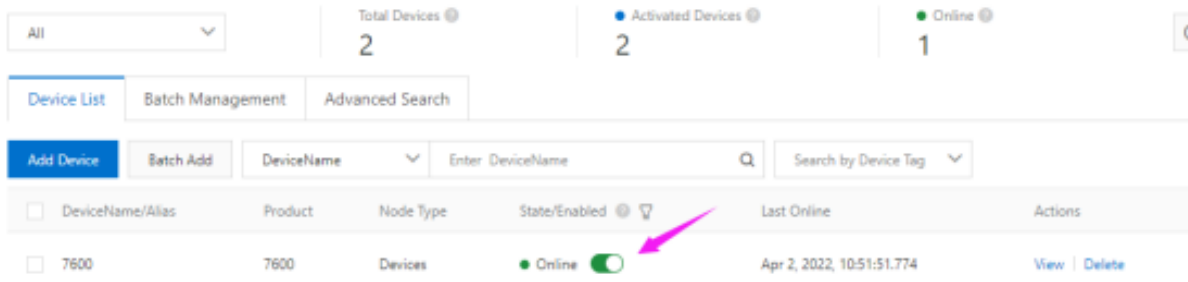
Flow Control: None

Get Default Save As Default Load Default Modify Key [firmware/Config](#) Restart Dev Modify Setting Cancel

(/wiki/File:RS485_TO_ETH_(B)_Connect07.png)

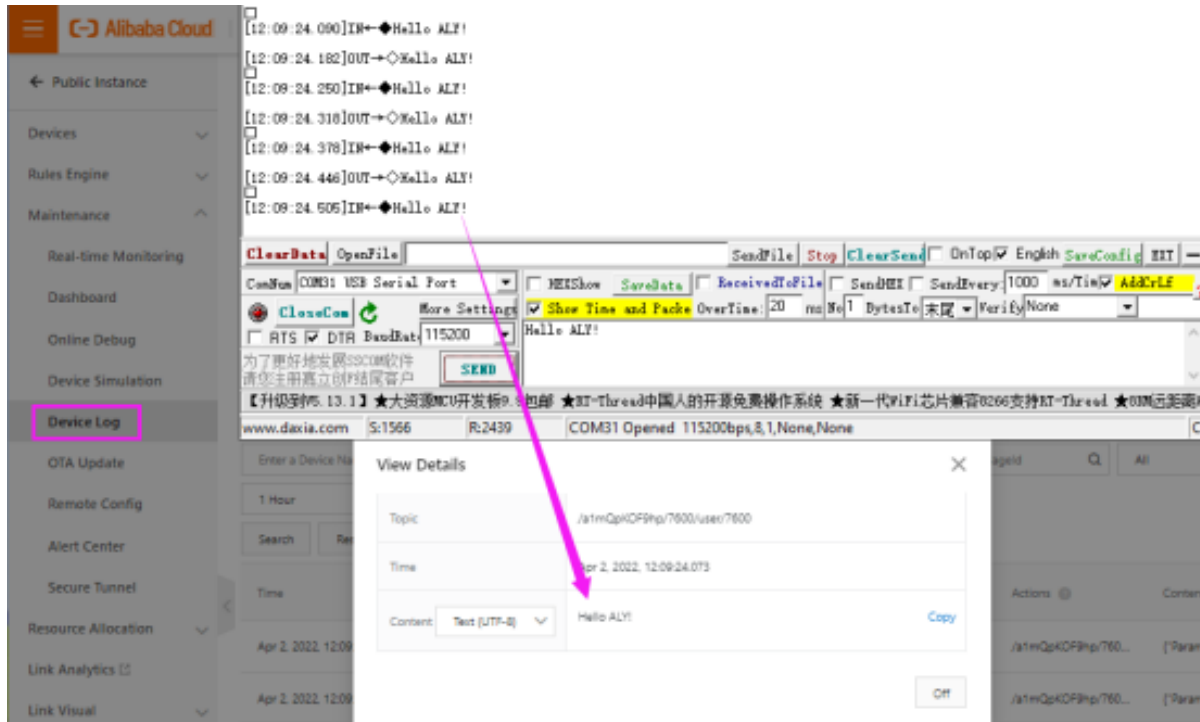
- IoT platform device display offline -> online.

Devices



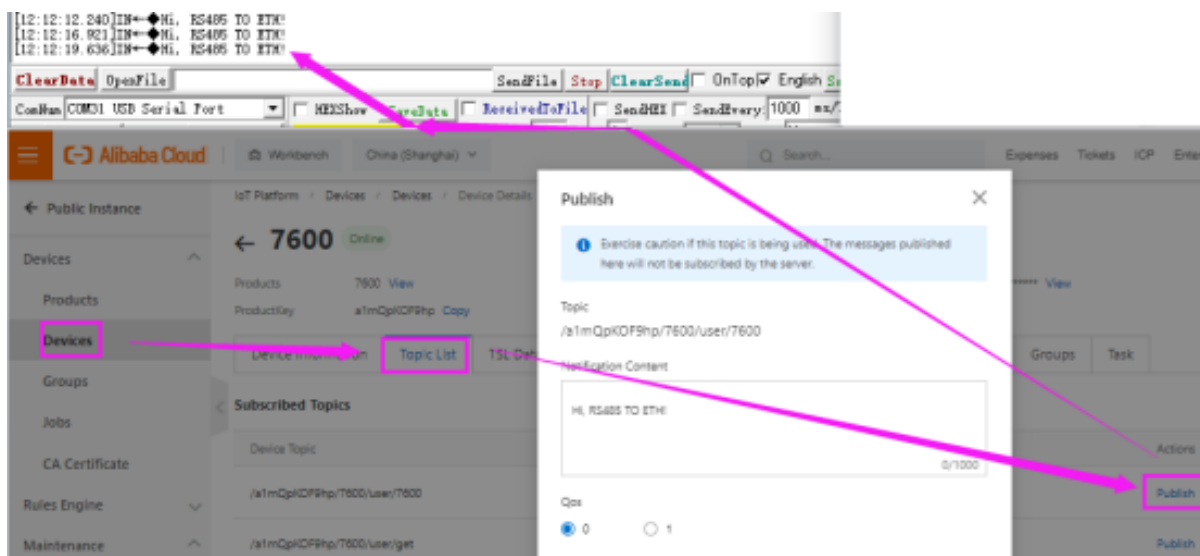
(/wiki/File:RS485_TO_ETH_(B)_Connect08.png)

- RS485 TO ETH(B) sends data to Alibaba Cloud MQTT server.



(/wiki/File:RS485_TO_ETH_(B)_Connect09.png)

- Alibaba Cloud IoT platform sends information to RS485 TO ETH(B).



(/wiki/File:RS485_TO_ETH_(B)_Connect10.png)

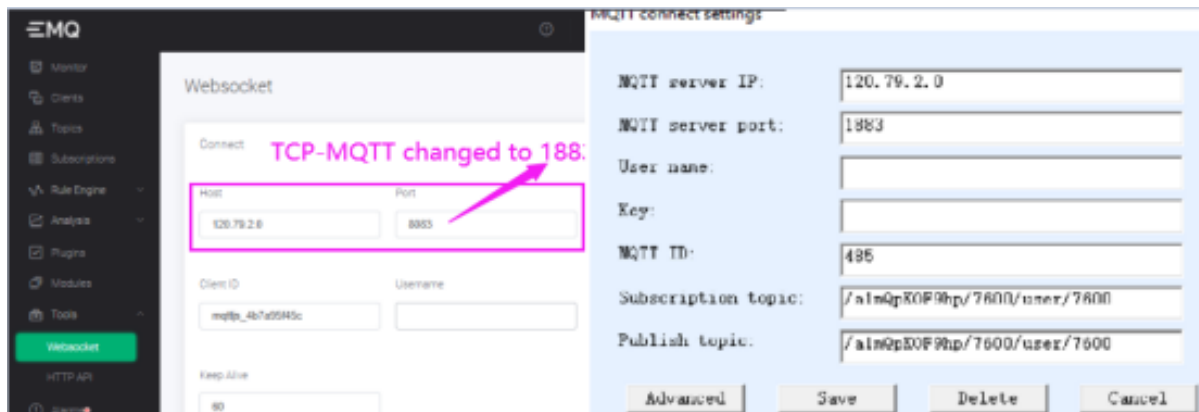
- RS485 TO ETH (B) Connect Alibaba Cloud and EMQX (optional).

Open Vircom->Device Management->Double-click the corresponding device->Firmware Config->MQTT Config.

Index	Type	Name	P.	Dev IP	Local ...	Dest IP	Work Mode	TCP Co...	Virtual Serial	Vircom Status	Dev ID	TXD	RXD
1	Subnet	WSDEV0001		192.168.10.61	0	47.103.184.100	TCP Client	Establis...	Haven't Bind	Not Linked	CAD56BAD	69013	25113
2	Subnet	WSDEV0001		192.168.10.66	0	120.79.2.0	TCP Client	Establis...	Haven't Bind	Not Linked	4F61F5AF	69264	24686

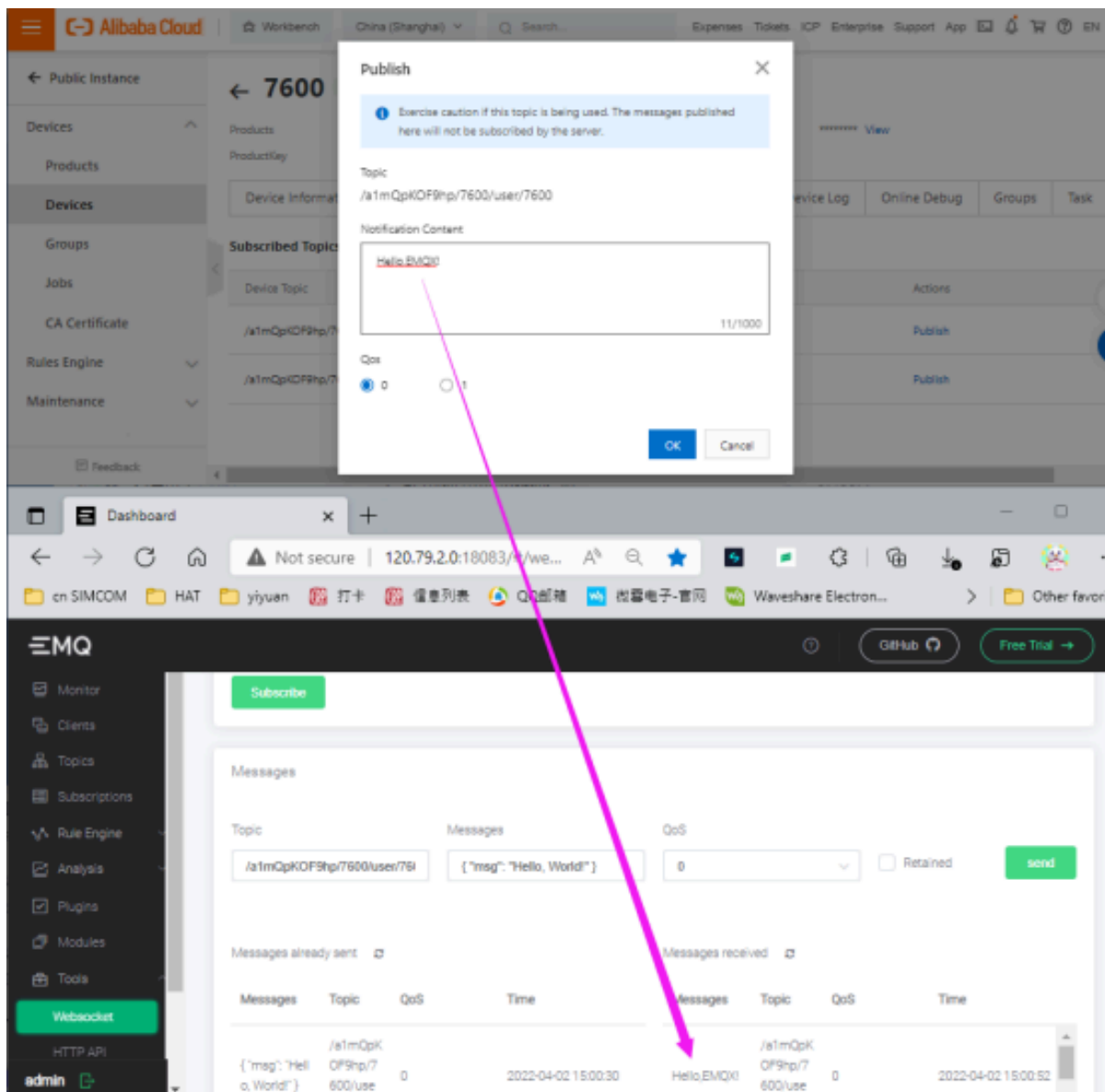
(/wiki/File:RS485_TO_ETH_(B)_Connect11.png)

- Configure EMQX's MQTT parameters to another RS485 TO ETH(B)



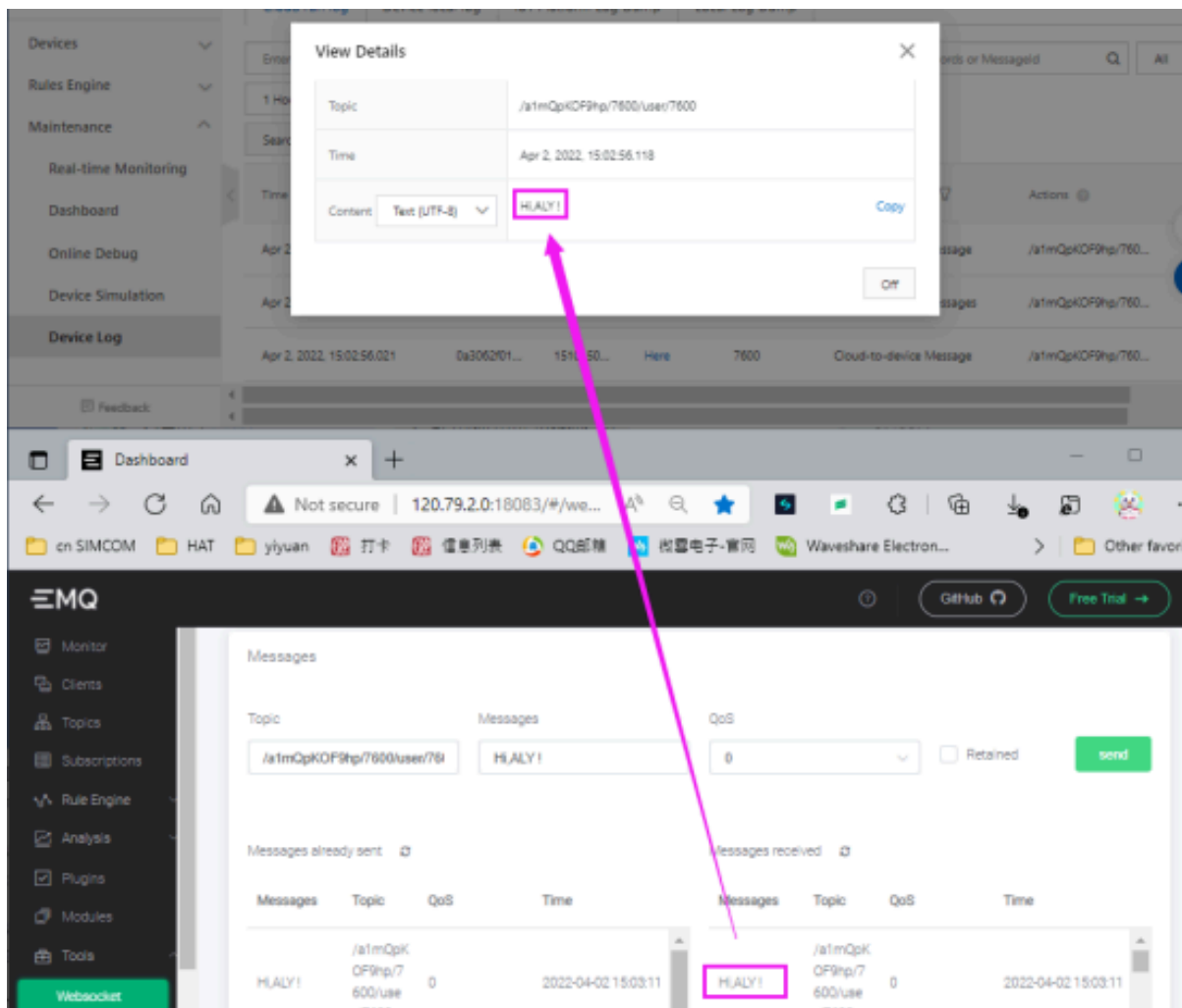
(/wiki/File:RS485_TO_ETH_(B)_Connect12.png)

- Alibaba Cloud sends data to EMQX: Hello, EMQX!



(/wiki/File:RS485_TO_ETH_(B)_Connect13.png)

- EMQX sends data to Alibaba Cloud: Hi, ALY!



(/wiki/File:RS485_TO_ETH_(B)_Connect14.png)

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([https://www.waveshare.com/w/index.php?title=RS485_TO_ETH_\(B\)_Connect_Alibaba_Cloud_And_EMQX&oldid=69174](https://www.waveshare.com/w/index.php?title=RS485_TO_ETH_(B)_Connect_Alibaba_Cloud_And_EMQX&oldid=69174))"